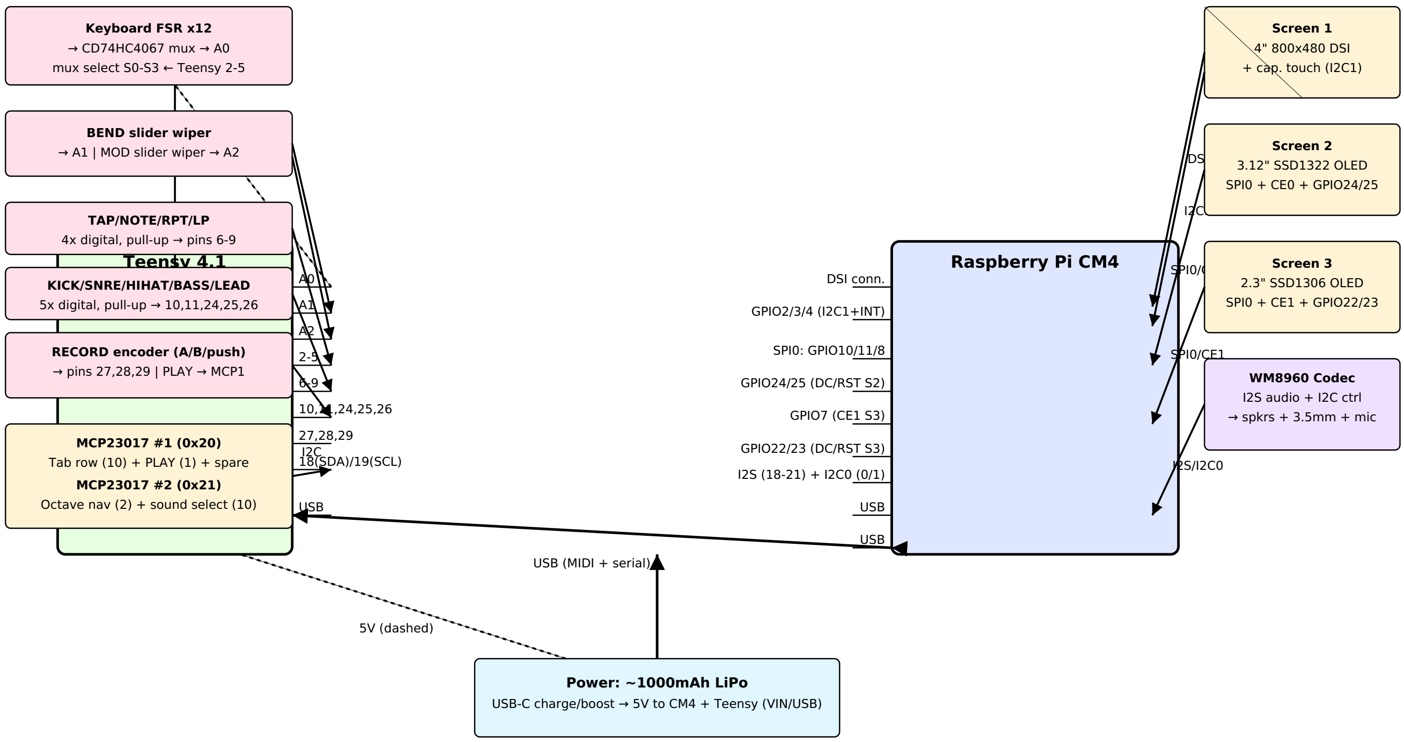
cy83rd3ck — PIN-LEVEL WIRING DIAGRAM



Reading the diagram: Solid arrows = direct wired connections with the pin/bus labeled. Dashed lines = secondary/shared connections (mux select lines, 5V power tap). The two MCP23017 I2C GPIO expanders sit on the same I2C bus as the Teensy (pins 18/19) and absorb the 22 low-priority buttons (tab row + octave/sound-select), keeping the Teensy's direct GPIO pins free for time-sensitive controls (sliders, transport buttons, encoder). CM4 GPIO numbers are BCM numbering. Exact MCP23017 pin-to-button mapping and the keyboard mux channel order would be defined in firmware/schematic next.

Package types: MCP23017 (x2, addr 0x20/0x21) — SSOP-28 SMD for final PCB, DIP-28 available for breadboard prototyping. CD74HC4067 16-ch analog mux (keyboard FSRs) — SOIC-24 SMD.

Connectors: USB-C — 16-pin SMD mid-mount receptacle (PD charging → IP5306). 3.5mm jack — slim PCB-mount SMD (PJ-320A-class, ~6mm depth). Screen 1 (DSI) — 22-pin 0.5mm-pitch FPC. Screens 2/3 (OLEDs) — 0.5mm-pitch FPC w/ matching ZIF sockets. Keyboard — ribbon/pitch TBD pending inspection of donor Raimy keybed PCB.